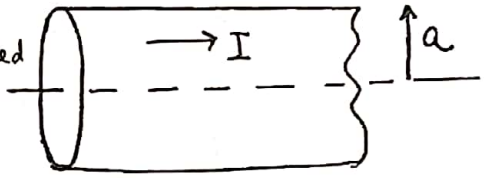
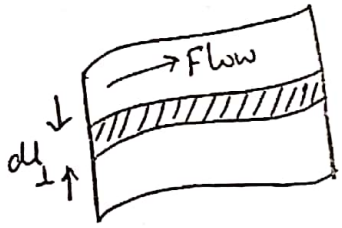
2.

(a) Current is uniformly distributed over the surface.



The surface current density is defined as $\vec{K} = \frac{d\vec{I}}{dl_\perp}$



where $dl_\perp$ is the length perpendicular to the flow of current

For a cylindrical wire of circular crossection in which current flows along the ~~direction~~ length of the cylinder, the total length perpendicular to the current flow is the circumference $2\pi a$

By definition, surface current of the wire is $K = \frac{I}{2\pi a}$

(b) It is given that $J \propto s$ or $J = k/s$ [s = distance from axis cylinder, k = constant]

By definition, $I = \oint \vec{J} \cdot \vec{da} = \oint J\, da_\perp$ [$da_\perp$ = magnitude of area perpendicular to the flow of current]

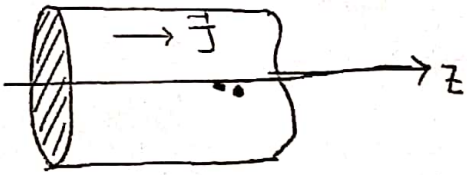
In cylindrical co-ordinates (s, φ, z)

$dl_s = ds$, $dl_\varphi = s\,d\varphi$, $dl_z = dz$

are infinitesimal lengths

We have to consider $\vec{da} = dl_s\, dl_\varphi (\hat{z})$

$= s\,ds\,d\varphi(\hat{z})$



$da = s\,ds\,d\varphi$ is the shaded area

$$I = \int_0^a \frac{k}{s}\, s\,ds \int_0^{2\pi} d\varphi = 2\pi k a \Rightarrow k = \frac{I}{2\pi a}$$

$$\therefore \boxed{J = \frac{k}{s} = \frac{I}{2\pi a s}}$$

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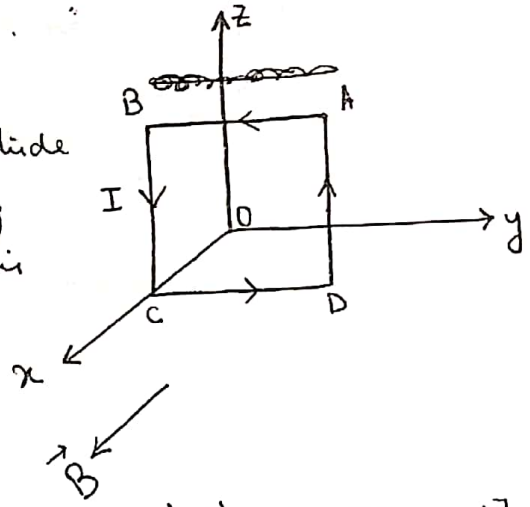
3.

$\vec{B} = kz\,\hat{x}$

A current of constant magnitude I flows anticlockwise along the square centered at O. Magnetic field is along x-direction



To calculate force use $\vec{F} = \int (\vec{I} \times \vec{B})\, dl = I \int (\vec{dl} \times \vec{B})$ [I = const]

Then direction of force is given by the direction $\vec{dl} \times \vec{B}$

$\vec{F}_{AB} = I\,[a(-\hat{y}) \times kz(\hat{x})] = Iakz\,\hat{z} = Ika \cdot \frac{a}{2}\,\hat{z} = \frac{Ika^2}{2}\,\hat{z}$

[$\because z = \frac{a}{2}$ for AB]

$\vec{F}_{CD} = I\,[a\hat{y} \times kz\,\hat{x}] = Iakz(-\hat{z})$. But $z = -\frac{a}{2}$ for CD

$= Iak(-\frac{a}{2}) \times (-\hat{z})$

$= Ik\frac{a^2}{2}(\hat{z})$

Now in similar manner $\vec{F}_{BC}$ and $\vec{F}_{DA}$ have same magnitudes but have opposite directions [$Iakz(-\hat{y})$ and $Iakz(\hat{y})$, respectively]

The net force along the upward direction is $(Ik\frac{a^2}{2} + Ik\frac{a^2}{2})\hat{z}$

$$\boxed{\vec{F}_{net} = Ika^2\,\hat{z}}$$

4. We know that magnetic field due to a current carrying straight wire is given by Biot-Savart law as:

$$B = \frac{\mu_0 I}{4\pi s}(\sin\theta_2 - \sin\theta_1) \quad \text{①}$$

where θ_1 and θ_2 are angles subtended at the field point (say P) by the finite ends of the wire. (done in class)

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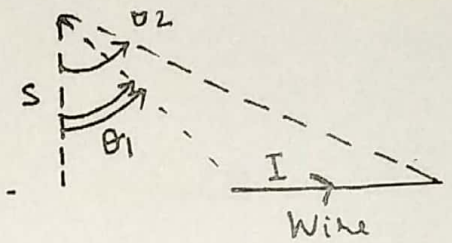
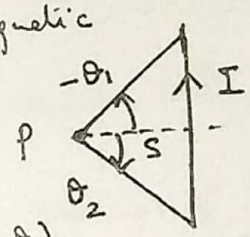
Note that in the expression above field point has been chosen such that θ_1 and θ_2 are both positive (same direction).



There can be field points (for e.g in the given problem) which subtend a +ve and a -ve value of θ, magnetic field at a distance s from the wire according to Biot Savart law would read

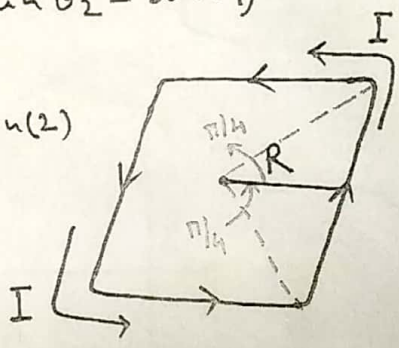


$$B = \frac{\mu_0 I}{4\pi s}\int_{-\theta_1}^{\theta_2} \cos\theta \, d\theta = \frac{\mu_0 I}{4\pi s}(\sin\theta_2 + \sin\theta_1) \quad \text{——— (2)}$$

In the above case, since θ_1 and θ_2 were both +ve

$$B = \frac{\mu_0 I}{4\pi s}\int_{\theta_1}^{\theta_2} \cos\theta \, d\theta = \frac{\mu_0 I}{4\pi s}(\sin\theta_2 - \sin\theta_1)$$

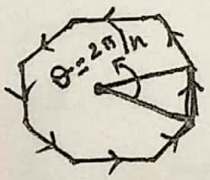
(a) In the given problem, using Eqn (2)



$$\left.\begin{aligned} B &= \frac{\mu_0 I}{4\pi R}\left(\sin\frac{\pi}{4} + \sin\frac{\pi}{4}\right) \\ \vec{B} &= \frac{\mu_0 I}{4\pi R}\left(\frac{2}{\sqrt{2}}\right)\hat{\varphi} \end{aligned}\right\} \text{ at center of loop}$$

Clearly for all 4 sides, $\vec{B}_{center} = \left(\frac{\mu_0 I}{4\pi R}\right)\frac{2}{\sqrt{2}} \times 4 \, \hat{\varphi} = \boxed{\sqrt{2}\,\frac{\mu_0 I}{\pi R}\,\hat{\varphi}}$

(b) Now instead of a square, I have a regular n-sided polygon – angle subtended by any side of the polygon at the center is $\frac{2\pi}{n}$. Put $\theta_1 = \theta_2 = \frac{\pi}{n}$ in Eqn (2)



$$B = \frac{\mu_0 I n}{4\pi R}\left[\sin\left(\frac{\pi}{n}\right) + \sin\left(\frac{\pi}{n}\right)\right] = \left(\frac{\mu_0 I n}{4\pi R}\right) 2\sin\left(\frac{\pi}{n}\right) = \frac{\mu_0 I}{2\pi R}\sin\left(\frac{\pi}{n}\right)$$

(n = no. of sides of polygon)

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(c) For $n \to \infty$, $\sin\left(\frac{\pi}{n}\right) \to \frac{\pi}{n}$

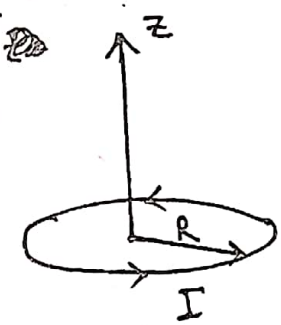
$$B = \frac{\mu_0 n I}{2\pi R} \frac{\pi}{n} = \frac{\mu_0 I}{2R} \quad \text{——(3)}$$

Now this field (3) is the same as the field at the center of a circular current-carrying loop.

It was shown in class that due to a circular loop, $\vec{B}$ is given by

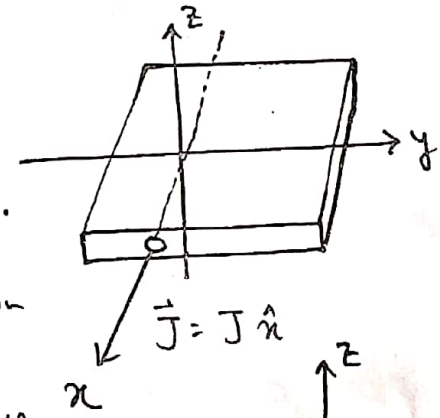
$$B = \frac{\mu_0}{4\pi} \frac{2\pi I R^2}{(R^2+z^2)^{3/2}}$$

[R: radius of the loo



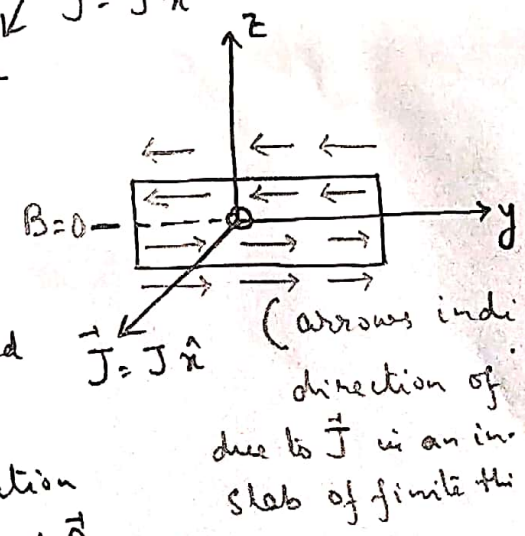
at center $R=0$, $B = \frac{\mu_0 I}{2R}$ which is same as (3)

5. Use Biot Savart law to determine the direction of the magnetic field $\vec{B}$ due to current density $\vec{J}$ and at point above and below $z=0$ plane.



$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{\vec{J} \times \hat{r}}{r^2} d\tau'$$

[$\hat{r}$: separation vector from volume current source and field point]



(arrows indi direction of due to $\vec{J}$ in an in slab of finite thi

Clearly $\vec{J} \times \hat{r}$ points along $(-\hat{y})$ for points above $z=0$ and along $(-\hat{y})$ for points below $z=0$. This can be verified by simple right hand rule.

Now that we have determined the direction of $\vec{B}$, we would find the magnitude of $\vec{B}$ by Ampere's circuital law.

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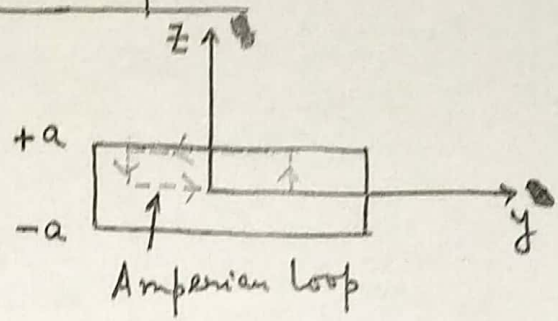
Now by symmetry $B = 0$ in $z = 0$ plane.

Inside the slab $0 < z < a$



Consider an amperian loop of length 'l' as shown inside the slab.

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc} \Rightarrow B.l = \mu_0 J \times z \times l \quad [B = 0 \text{ at } z = 0]$$

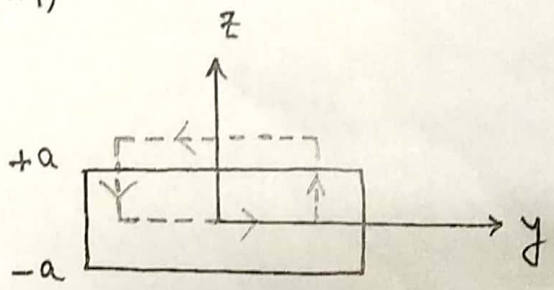
(Ampere's law) $\Rightarrow B = \mu_0 J z$ (magnitude of $\vec{B}$)

Considering directions from Biot-Savart law, we can write

$$\boxed{\vec{B}_{in} = -\mu_0 J z (\hat{y})} \quad (-a < z < a)$$

$$\cancel{= -\mu_0 J a (\hat{y}) \quad [z > 0]}$$
$$\cancel{= +\mu_0 J a (\hat{y}) \quad [z < 0]}$$

Outside the slab $(z > |a|)$



Consider again amperian loop of length l.

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$$

$$Bl = \mu_0 J \times a \times l \quad [I_{enc} \text{ this time is } J \times a \times l]$$

$$B = \mu_0 J a$$

$$\therefore \boxed{\vec{B}_{out} = -\mu_0 J a (\hat{y})}$$

$$\vec{B}_{out} = -\mu_0 J a \hat{y} \quad z > a$$
$$= +\mu_0 J a \hat{y} \quad z > -a$$

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